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Photoresist Development

The chemistry of photoresists is designed to turn a spatial distribution of energy (the aerial image) into a spatial distribution of resist solubility. Ultimately, the dissolution process turns the continuous energy distribution of the projected aerial image into a binary resist image: either the resist is dissolved or it remains on the wafer. Exposure and post-exposure bake create latent images of chemical distributions, but it is the chemistry of dissolution that has the greatest impact on the ability to discern between line and space and to control the dimensions of the final resist features.

A kinetic approach to understanding development will be taken here, with a postulated reaction mechanism that then leads to a development rate equation. Deviations from the expected kinetic development rates at the surface of the resist, called surface induction or surface inhibition, will be related empirically to the expected development rate, i.e. to the bulk development rate. Next, the influence of developer temperature and normality are discussed. Once a basic understanding of what affects the development rate is established, the traditional resist metric ‘contrast’ will be defined and explored. Finally, to arrive at a resist profile shape, the physical mechanism by which a spatial variation in development rate turns into a final resist profile is examined by way of the development path.

7.1 Kinetics of Development

In order to derive an analytical development rate expression, a simple kinetic model of the development process will be used. This approach involves proposing a reasonable mechanism for the development reaction and then applying standard kinetics to this mechanism in order to derive a rate equation.

Unfortunately, fundamental experimental evidence of the exact mechanism of photoresist development is lacking. The models presented below are reasonable, and the resulting rate equations have been shown to describe actual development rates extremely well.

However, faith in the exact details of the mechanism is limited by this dearth of fundamental studies. In addition, photoresists are not simple, uniform collections of model compounds. Distributions of polymers with different chain lengths (molecular weights) and functional groups mean that macroscopic observations reflect average material properties that are only approximately mapped to microscopic mechanisms.

7.1.1 A Simple Kinetic Development Model

Our discussion here will apply to both conventional and chemically amplified positive photoresists, and will be generalized to negative working resists in a later section. For all resist systems in common use today, the developer is a dilute aqueous solution of a base, most commonly tetramethyl ammonium hydroxide (TMAH).

Photoresist dissolution involves three processes: diffusion of the active component of the developer from the bulk solution to the surface of the resist, reaction of the developer with the resist, and diffusion of the product back into the solution. For this analysis, we shall assume that the last step, diffusion of the dissolved resist into solution, occurs very quickly so that this step may be ignored. Let us now look at the first two steps in the proposed mechanism. The diffusion of developer to the resist surface can be described with the simple diffusion rate equation, given approximately by a simple difference equation:

$$r_D = k_D(D - D_s) \quad (7.1)$$

where r_D is the rate of diffusion of the developer to the resist surface, D is the bulk developer concentration, D_s is the developer concentration at the resist surface, and k_D is the rate constant (equal to the diffusivity of the developer in the solution divided by the boundary layer thickness).

We shall now propose a mechanism for the reaction of developer with the resist. It is quite likely that this step is in fact a series of more detailed steps, including the diffusion of the developer cation into the resist to form a thin gel layer. However, we will assume a simple surface-limited reaction here. The resist is composed of large macromolecules of resin R along with a dissolution inhibitor M . For the conventional DNQ/novolac resist, the dissolution inhibitor is the photoactive compound itself. For a chemically amplified resist, dissolution is inhibited by the blocking group attached to the PHS (polyhydroxy styrene) resin. The resin is somewhat soluble in the developer solution, but the presence of the inhibitor makes the development rate very slow. Solubility in the resist is changed by removing the inhibitor. For a conventional resist, the sensitizer converts to carboxylic acid upon exposure to UV light. This carboxylic acid is highly soluble in developer, enhancing the dissolution rate of the resin. For a chemically amplified resist, the PEB amplification reaction deblocks the polymer, generating the highly soluble $-OH$ site.

Let us assume that n inhibitor sites must be removed from the influence of one resin in order for the developer to dissolve that resin molecule. This n parameter, the reaction order for this step in the mechanism, will turn out to be the most critical development parameter, representing the number of exposure or deblocking events that work together to get one resin molecule into solution. The rate of the reaction for this step is

$$r_R = k_R D_s (M_0 - M)^n \quad (7.2)$$

where r_R is the rate of reaction of the developer with the resist, k_R is the rate constant, and M_0 is the initial inhibitor concentration. The two steps outlined above are in series, i.e. one reaction follows the other. Thus, the two steps will come to a steady state such that

$$r_R = r_D = r \quad (7.3)$$

Equating the rate equations, one can solve for D_S and eliminate it from the overall rate equation, giving the steady-state development rate:

$$r = \frac{k_D k_R D (M_0 - M)^n}{k_D + k_R (M_0 - M)^n} \quad (7.4)$$

Letting $m = M/M_0$, the relative inhibitor concentration, Equation (7.4) becomes

$$r = \frac{k_D D (1 - m)^n}{k_D/k_R M_0^n + (1 - m)^n} \quad (7.5)$$

When $m = 1$ (resist unexposed/unreacted), the rate is zero. When $m = 0$ (resist completely exposed/reacted), the rate is equal to $r_{\max}$ where

$$r_{\max} = \frac{k_D D}{k_D/k_R M_0^n + 1} \quad (7.6)$$

If we define a constant a such that

$$a = k_D/k_R M_0^n \quad (7.7)$$

the rate equation becomes

$$r = r_{\max} \frac{(a + 1)(1 - m)^n}{a + (1 - m)^n} \quad (7.8)$$

Note that the simplifying constant a describes the rate constant of diffusion relative to the surface reaction rate constant. A large value of a will mean that diffusion is very fast, and thus less important, compared to the fastest surface reaction (for completely exposed/reacted resist), and the rate is reaction-controlled. When a is small, the rate is diffusion-controlled.

There are three constants that must be determined experimentally: a , n and $r_{\max}$. The constant a can be put in a more physically meaningful form as follows. A characteristic of some experimental rate data is an inflection point in the rate curve at values of m between 0.2 and 0.8. The point of inflection can be calculated by letting

$$\frac{d^2 r}{dm^2} = 0 \quad \text{at} \quad m = m_{th}$$

giving

$$a = \frac{(n + 1)}{(n - 1)} (1 - m_{th})^n \quad (7.9)$$

where m_{th} is the value of m at the inflection point, called the *threshold inhibitor concentration*.

This model so far does not take into account the nonzero dissolution rate of unexposed resist (r_{min}). Simply adding this term to Equation (7.8) gives what has been called the ‘*original*’ kinetic model or the ‘Original Mack model’:¹

$$r = r_{max} \frac{(a+1)(1-m)^n}{a + (1-m)^n} + r_{min} \quad (7.10)$$

This approach assumes that the mechanism of development of the unexposed resist is independent of the above-proposed development mechanism. In other words, there is a finite dissolution of resist that occurs by a mechanism that is independent of the presence of exposed photoactive compound (PAC). Note that the addition of the r_{min} term means that the true maximum development rate is actually $r_{max} + r_{min}$. In most cases $r_{max} \gg r_{min}$ and the difference is negligible.

Consider the case when the diffusion rate constant is large compared to the surface reaction rate constant (i.e. the rate is reaction-controlled). If $a \gg 1$, the development rate Equation (7.10) will become

$$r = r_{max}(1-m)^n + r_{min} \quad (7.11)$$

The interpretation of a as a function of the threshold inhibitor concentration m_{th} given by Equation (7.9) means that a very large a would correspond to a large negative value of m_{th} . In other words, if the surface reaction is very slow compared to the mass transport of developer to the surface there will be no inflection point in the development rate data and Equation (7.11) will apply. It is quite apparent that Equation (7.11) could be derived directly from Equation (7.2) if the diffusion step were ignored.

Figure 7.1 shows some plots of this model for different values of n . The behavior of the dissolution rate with increasing n values is to make the rate function more ‘selective’

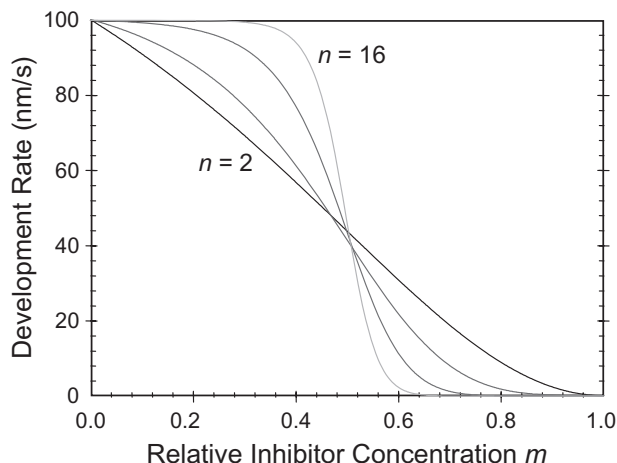


Figure 7.1 Development rate plot of the original kinetic model as a function of the dissolution selectivity parameter ($r_{max} = 100 \text{ nm/s}$, $r_{min} = 0.1 \text{ nm/s}$, $m_{th} = 0.5$, and $n = 2, 4, 8$ and 16)

between resist exposed/reacted above m_{th} and resist exposed/reacted below m_{th} . For this reason, n is called the *dissolution selectivity parameter*. Also from this behavior, the interpretation of m_{th} as a ‘threshold’ concentration becomes quite evident.

The key parameter in this development model that affects lithographic performance is the dissolution selectivity n . As n goes to infinity, the behavior seen in Figure 7.1 reaches its extreme of a step function. In other words, for $n = \infty$ the resist becomes an ideal threshold resist. Physically, this parameter represents the number of exposure or deblocking reactions that work together to cause one resin molecule to go into solution. Engineering the materials of the resist to increase n is one of the primary tasks of the resist designer. In general, a high n implies a high molecular weight of the resin (for example, a large number of deblocking reactions per resin means a large chain length of the resin). Unfortunately, a resist with all high molecular weight resin will tend to develop more slowly (lower r_{max} and m_{th}), thus degrading resist sensitivity.

The second most important factor in resist performance is the ratio of the maximum to minimum development rates, r_{max}/r_{min} . Increasing this ratio above 100 will definitely improve lithographic performance, though there will be diminishing returns as the ratio exceeds 10000 or so.

7.1.2 Other Development Models

The previous kinetic model is based on the principle of dissolution enhancement. For a diazonaphthoquinone (DNQ)/novolac resist, the exposure product enhances the dissolution rate of the resin/PAC mixture. For a chemically amplified resist, the deblocked site enhances the solubility of the blocked polymer. In reality this is a simplification – there are really two mechanisms at work. The inhibitor acts to inhibit dissolution of the resin while the product that results from removing that inhibitor enhances dissolution. Thus, a development rate expression could reflect both of these mechanisms. A new model, call the *enhanced kinetic model*, has been proposed to include both effects:²

$$r = r_{resin} \frac{1 + k_{enh}(1 - m)^n}{1 + k_{inh}(m)^l} \quad (7.12)$$

where k_{enh} is the rate constant for the enhancement mechanism, n is the enhancement reaction order, k_{inh} is the rate constant for the inhibition mechanism, l is the inhibition reaction order, and r_{resin} is the development rate of the resin alone.

For no exposure/reaction, $m = 1$ and the development rate is at its minimum. From Equation (7.12),

$$r_{min} = \frac{r_{resin}}{1 + k_{inh}} \quad (7.13)$$

Similarly, when $m = 0$, corresponding to complete exposure/reaction, the development is at its maximum.

$$r_{max} = r_{resin}(1 + k_{enh}) \quad (7.14)$$

Thus, the development rate expression can be characterized by five parameters: r_{max} , r_{min} , r_{resin} , n and l .

The enhanced kinetic model for resist dissolution is a superset of the reaction-controlled version of the original kinetic model. If the inhibition mechanism is not important, then $k_{\text{inh}} = 0$. For this case, Equation (7.12) is identical to Equation (7.11) when

$$r_{\min} = r_{\text{resin}}, \quad r_{\max} = r_{\text{resin}} k_{\text{enh}} \quad (7.15)$$

The enhanced kinetic model of Equation (7.12) assumes that mass transport of developer to the resist surface is not significant. Of course, a simple diffusion of developer can be added to this mechanism as was done above with the original kinetic model (see Problem 7.3).

Figure 7.2 shows several plots of this model. The five parameters of the enhanced kinetic model give this equation more flexibility in fitting experimental development rate curves compared to the four-parameter original kinetic model. As Figure 7.2 shows, the combination of enhancing and inhibiting mechanisms can produce a sort of double threshold behavior with rates rising and lowering around a plateau at the resin development rate.

Perhaps a more intuitive way to understand the basic mechanism of the enhanced kinetic rate model is with the Meyerhofer plot.³ This plot, such as the one shown in Figure 7.3, shows how the addition of increasing concentrations of inhibitor in the resist formulation (in this case DNQ added to a novolac resin) changes the original development rate of the resin and increases $r_{\max}$ while decreasing $r_{\min}$. Experimentally, the Meyerhofer plot shows an approximately linear dependence of log-development rate on the initial concentration of inhibitor, M_0 (Figure 7.3a). This empirical result can be explained reasonably well if one assumes that k_{enh} has a first-order dependence on M_0 and k_{inh} has a second-order dependence on M_0 , as seen in Figure 7.3b.

Some experimentally derived $r(m)$ behavior is not well fit by either kinetic model described above. Figure 7.4 shows the best fit of both the original and the enhanced development rate models to one set of data.⁴ In the region around a 0.5 inhibitor concentration, the data exhibits a ‘notch’ behavior where the actual development rate drops very quickly to a value much less than that predicted by either model. This sudden drop in

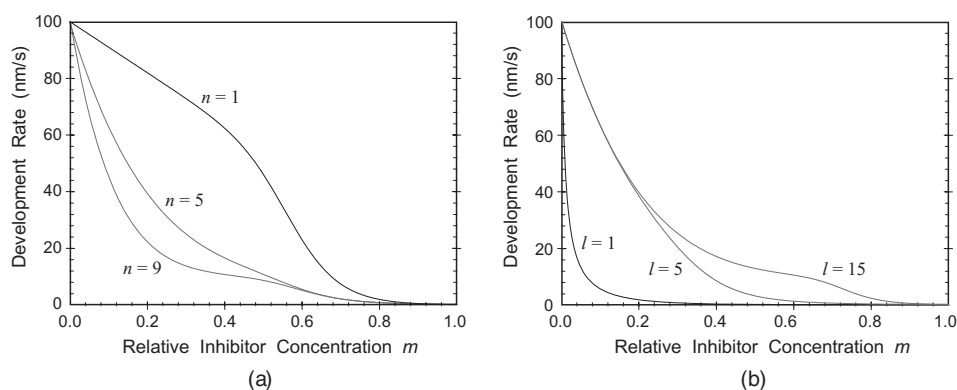


Figure 7.2 Plots of the enhanced kinetic development model for $r_{\max} = 100 \text{ nm/s}$, $r_{\text{resin}} = 10 \text{ nm/s}$, $r_{\min} = 0.1 \text{ nm/s}$ with: (a) $l = 9$; and (b) $n = 5$

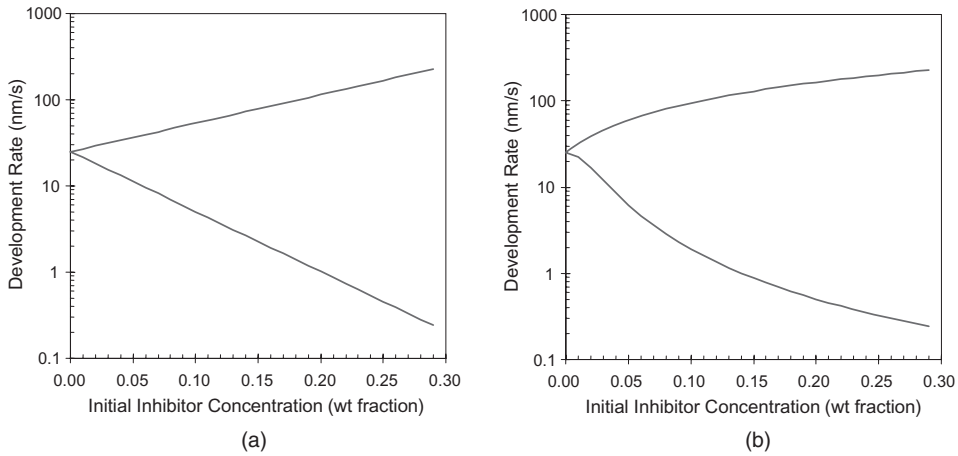


Figure 7.3 An example of a Meyerhofer plot, showing how the addition of increasing concentrations of inhibitor increases $r_{\max}$ and decreases $r_{\min}$: (a) idealized plot showing a log-linear dependence on initial inhibitor concentration, and (b) the enhanced kinetic model assuming $k_{\text{enh}} \propto M_0$ and $k_{\text{inh}} \propto M_0^2$

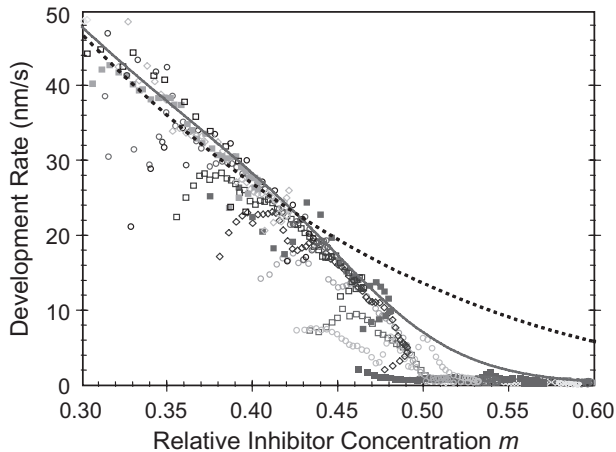


Figure 7.4 Comparison of experimental dissolution rate data (symbols) exhibiting the so-called 'notch' shape to best fits of the original (dotted line) and enhanced (solid line) kinetic models. The data shows a steeper drop in development rate at about 0.5 relative inhibitor concentration than either model predicts

rate, resembling a notch in the otherwise slowly varying behavior, is critical to resist performance because it is the region of steepest change in relative development rate versus m . Thus, it is critical that a development model properly describes this region of the dissolution rate curve in order to accurately predict the behavior of the resist.

A semiempirical model has been devised in order to better fit the notch behavior described above. This *notch model* begins with the simple version of the original model

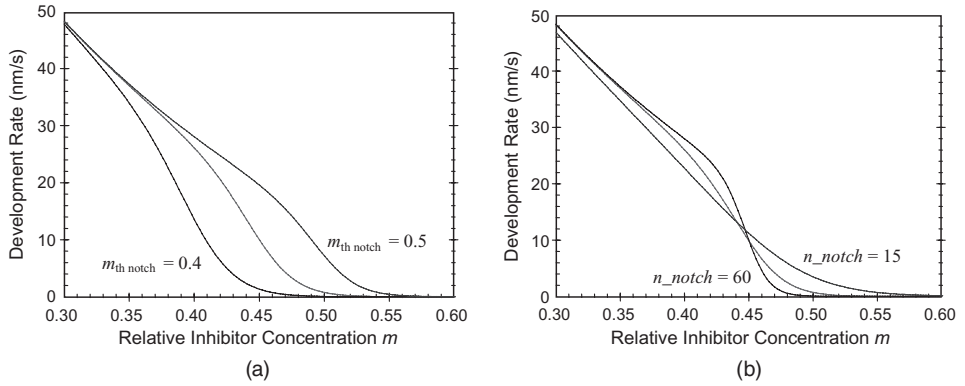


Figure 7.5 Plots of the notch model: (a) m_{th_notch} equal to 0.4, 0.45 and 0.5 with n_notch equal to 30; and (b) n_notch equal to 15, 30 and 60 with m_{th_notch} equal to 0.45

given in Equation (7.11) and adds a notch function equivalent to the threshold behavior given by Equation (7.10):

$$r = r_{\max}(1-m)^n \left[\frac{(a+1)(1-m)^{n_notch}}{a + (1-m)^{n_notch}} \right] + r_{\min} \quad (7.16)$$

where

$$a = \frac{(n_notch + 1)}{(n_notch - 1)} (1 - m_{TH_notch})^{n_notch}$$

The term in the square brackets of Equation (7.16) provides the notch-like behavior where m_{th_notch} is the position of the notch along the inhibitor concentration axis and n_notch gives the strength of the notch. This is illustrated in Figure 7.5. Note that the five-parameter notch model of Equation (7.16) reverts to the original kinetic model of Equation (7.10) when $n = 0$ and becomes equivalent to the simplified kinetic model of Equation (7.11) when $n_notch = 1$.

Of the three models described above, about half of the resists that have been characterized are fit well with the original kinetic model. Most of the rest are well described by the enhanced kinetic model, with only a few resists needing the notch model to accurately describe their dissolution properties.

7.1.3 Molecular Weight Distributions and the Critical Ionization Model

In each of the models above, the mechanism of surface reaction is based on the idea that n dissolution-changing chemical events are required to make one polymer molecule dissolve. Real resists, however, are formulated with polymers that have a range of molecular weights (that is, different polymer chain lengths). It seems unlikely that each polymer molecule, regardless of its chain length, would require the same number of deblocking events (taking chemically amplified resists as an example) to make it soluble. Another approach, called the *critical ionization model*, assumes that the polymer becomes soluble

whenever some critical fraction of its blocked groups become deblocked.⁵ Thus, lower molecular weight polymers would require fewer deblocking events (lower n) to become soluble compared to higher molecular weight polymers.

High molecular weight polymers (>5000 g/mol, for example) result in high n , but with less than ideal sensitivity. By mixing in a small amount of low molecular weight polymer (<500 g/mol, for example),⁶ resist sensitivity can be improved without a significant reduction in average n . Mid-molecular weight resins, however, tend to degrade resolution by lowering n .

7.1.4 Surface Inhibition

The kinetic models given above predict the development rate of the resist as a function of the inhibitor concentration remaining after the resist has been exposed/reacted. There are, however, other parameters that are known to affect the development rate, but that were not included in these models. The most notable deviation from the kinetic theory is the surface inhibition effect (Figure 7.6). The inhibition, or surface induction, effect is a decrease in the expected development rate at the surface of the resist.⁷ Thus, this effect is a function of the depth into the resist and requires a new description of development rate.

Several factors have been found to contribute to the surface inhibition effect. Baking of the photoresist can produce surface inhibition and two possible mechanisms are thought to be likely causes. One possibility is an oxidation of the resist at the resist surface, resulting in reduced development rate of the oxidized film. Alternatively, the induction effect may be the result of reduced solvent content near the resist surface, which also results from baking the resist (see Chapter 5). Both mechanisms could be contributing to the surface inhibition. Quite commonly, surface inhibition can be induced with the use of surfactants (surface-acting agents) in the developer. Chemically amplified resists open a wide array of mechanisms for surface inhibition due to reduced amplification near the

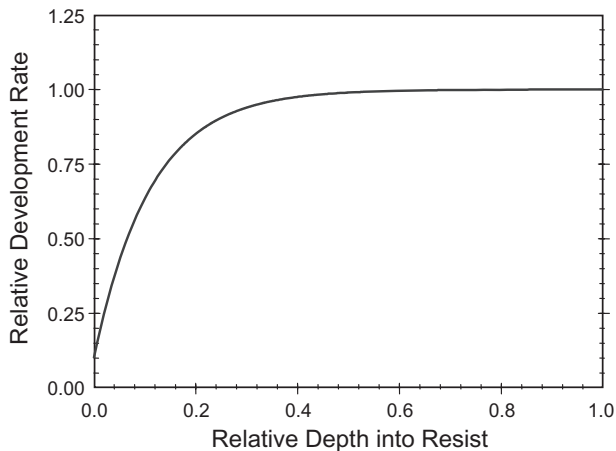


Figure 7.6 Example surface inhibition with $r_0 = 0.1$ and $\delta = 100\text{ nm}$ for a 1000-nm-thick resist

resist surface (caused, for example, by acid evaporation, base contamination, or reduced acid diffusion in the low-solvent surface region).

An empirical model can be used to describe the positional dependence of the development rate. If we assume that the development rate near the surface of the resist exponentially approaches the bulk development rate, the rate as a function of depth, $r(z)$, is

$$r(z) = r_B[1 - (1 - r_0)e^{-z/\delta}] \quad (7.17)$$

where r_B is the bulk development rate, r_0 is the development rate at the surface of the resist relative to r_B , and δ is the depth of the surface inhibition layer. In several conventional resists, the induction effect has been found to take place over a depth of about 100 nm or less.⁸ For chemically amplified resists, 200- to 300-nm inhibition depths can be observed.

A small amount of surface inhibition can be good for photoresist profile control. Slower development of the top of the resist can lead both to less resist loss for small lines, and also profiles with sharper, more square tops. Too much inhibition, however, can lead to T-top-shaped profiles and a loss of linewidth control. In general, moderate values of r_0 and large values of δ lead to improved resist profiles without significant loss of dimensional control.

One of the most useful aspects of surface inhibition is when this effect is used to counter the decrease in development rate caused by absorption in positive resists. For resist sitting on a good bottom antireflection coating, the energy exposing the resist falls off exponentially with depth into the resist. We can approximate the impact of this absorption on the variation of development rate with depth into the resist in the following way. For a conventional resist, ignoring resist bleaching and assuming small exposure doses, the sensitizer will vary with depth into the resist approximately as

$$m(z) = \exp[-CDose_{inc}I_r(z)] \approx 1 - CDose_{inc}I_r(z) = 1 - CDose_{inc}e^{-\alpha z} \quad (7.18)$$

Using the simple reaction-limited development rate Equation (7.11) and ignoring the minimum development rate,

$$r(z) \approx r_{max}(1 - m(z))^n \approx r(z=0)e^{-\alpha n z} \quad (7.19)$$

Thus, the effect of absorption is to cause something close to an exponentially decaying development rate with depth into the resist. Surface inhibition can counteract some of this falloff. Reasonable results are obtained when the surface development rate is set to match the development rate at the bottom of the resist and the surface inhibition depth is set to match the absorption exponential decay depth.

$$r_0 \approx e^{-\alpha n D}, \quad \delta \approx \frac{1}{\alpha n} \quad (7.20)$$

where D is the resist thickness.

Note that the observed magnitude of surface effects is often on the order of 50–100 nm. Thus, for thin resist process, where the thickness of the resist is on the order of 100 nm, the very idea that resist response can be separated into surface and bulk behavior is called into question.

7.1.5 Extension to Negative Resists

The development rate models discussed above were derived based on positive resist chemistry. However, with only a slight change in interpretation, these models are perfectly applicable to most negative resist chemistries. For the three models given by Equations (7.10), (7.12) and (7.16), the term m represents the concentration of the species that inhibits dissolution. For a conventional positive resist, this quantity is the photoactive compound. For conventional negative resists, the inhibitor is the exposure product. By interpreting the m in the development rate equations as the inhibitor concentration for the negative resist, whatever it might be, the rate equations can be used without further modification.

7.1.6 Developer Temperature

It is well known that the temperature of the developer solution during development can have a significant impact on resist performance. The speed (i.e. overall development rate) varies in a complicated way with temperature, usually resulting in the counterintuitive result of a 'faster' resist process (i.e. a process requiring lower exposure doses) at lower developer temperatures. The shape of the development rate versus dose (or versus inhibitor concentration) curve will also vary considerably with temperature, leading to possibly significant performance differences. The effect of temperature on dissolution rate shows a complicated behavior where changes in developer temperature give changes in dissolution rate that are dose dependent. At one dose, the effect of temperature on dissolution rate can be very different than at another dose.⁹ Use of a dissolution rate model can simplify the description of temperature effects by showing just the change in the model parameters with developer temperature.

Figure 7.7 shows an example of how temperature can affect dissolution rate. In general, one usually expects simple kinetic rate-limited reactions to proceed faster at higher

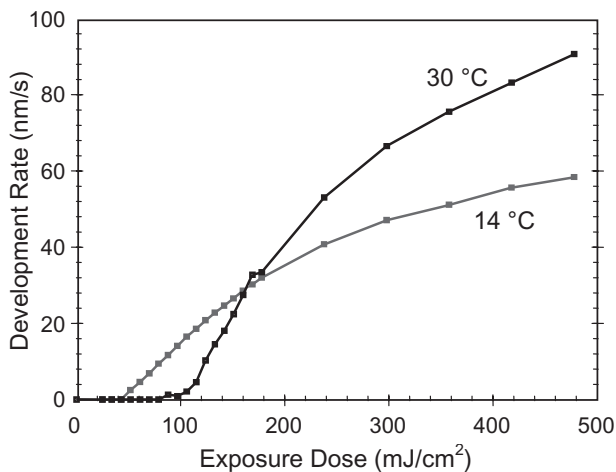


Figure 7.7 Development rate of THMR-iP3650 (averaged through the middle 20 % of the resist thickness) as a function of exposure dose for different developer temperatures shows a change in the shape of the development dose response

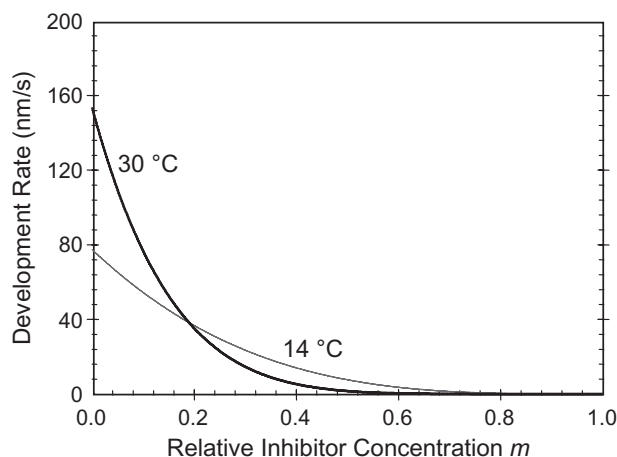


Figure 7.8 Comparison of the best-fit models of THMR-iP3650 for different developer temperatures shows the effect of increasing $r_{\max}$ and increasing dissolution selectivity parameter n on the shape of the development rate curve

temperatures (indicating a positive activation energy for the reaction). The behavior shown in Figure 7.7 is obviously more complicated than that. At high doses, increasing developer temperature does increase the development rate. But at low doses the opposite is true. Thus, developer temperature has a significant impact on the shape of the dissolution rate curve. Reasons have been proposed for this behavior,¹⁰ but here we will only strive to accurately describe this behavior quantitatively using model parameters.

By fitting the dissolution rate behavior to a development model, the variation of the $r(m)$ curve with temperature can be deduced, as shown in Figure 7.8. Again, the results show that at high doses (corresponding to low concentrations of inhibitor remaining), higher developer temperature increases the development rate. But at low doses (high concentrations of inhibitor remaining), the opposite is true. Using the terminology of the original kinetic development model, increasing the developer temperature caused an increase in the maximum development rate $r_{\max}$ and an increase in the dissolution selectivity parameter n .

Figure 7.9 shows the final results of the analysis. The two parameters $r_{\max}$ and n are plotted versus developer temperature in an Arrhenius plot. For the i-line resist THMR-iP3650, the activation energies for $r_{\max}$ and n are 7.41 and 7.02 Kcal/mole, respectively.¹¹

Not all resists exhibit the simple, regular behavior with developer temperature described above. The chemically amplified resist Apex-E, for example, shows a maximum dissolution n value at about 35 °C, as seen in the data of Table 7.1. More work is required to fully understand the impact of developer temperature on a wider array of resists.

7.1.7 Developer Normality

TMAH developer at 0.26N (2.38 % by weight) is extremely widely used (almost to the point of being a universal standard). However, as one would expect, developer normality

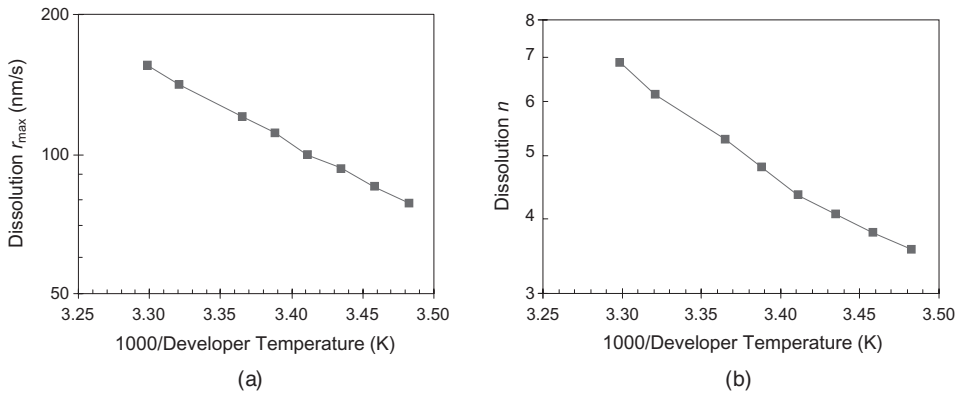


Figure 7.9 Arrhenius plots of (a) the maximum dissolution rate $r_{\max}$ and (b) the dissolution selectivity parameter n for THMR-iP3650

Table 7.1 Developer temperature-dependent modeling parameters of Apex-E (0.26 N developer)¹²

Developer Temperature (°C)	$r_{\max}$ (nm/s)	$r_{\min}$ (nm/s)	m_{th}	n
5	53.1 ± 9.0	1.18	0.571 ± 0.036	3.93 ± 0.56
10	68.0 ± 10.9	1.283	0.578 ± 0.031	4.17 ± 0.52
15	91.9 ± 13.8	1.35	0.586 ± 0.028	4.25 ± 0.48
20	115.7 ± 14.1	1.48	0.682 ± 0.016	4.79 ± 0.49
25	146.1 ± 15.2	1.55	0.637 ± 0.015	5.56 ± 0.54
30	177.8 ± 16.0	1.62	0.646 ± 0.012	6.36 ± 0.55
35	199.7 ± 14.6	2.56	0.693 ± 0.008	15.58 ± 2.17
40	196.2 ± 13.1	2.14	0.694 ± 0.008	11.25 ± 1.32
45	209.5 ± 22.5	1.83	0.665 ± 0.014	6.64 ± 0.74

(the concentration of TMAH) impacts dissolution rates greatly. In general, higher normality produces higher dissolution rates. Using again the model coefficients as a way of describing the impact of developer normality on dissolution, Equation (7.6) would suggest that the parameters $r_{\max}$ and $r_{\min}$ increase with an increase concentration of TMAH. The important dissolution selectivity parameter n tends to have an optimum, essentially going to zero at very high and very low normalities. The original kinetic model parameters $r_{\max}$ and the dissolution selectivity parameter n vary with developer normality as shown in Table 7.2 for the chemically amplified resist Apex-E. As the developer temperature increases, the impact of different developer normalities on $r_{\max}$ is lessened. Although the data here is limited, $r_{\max}$ exhibits a well-known phenomenon – there is a critical normality (between about 0.1 and 0.15 N) below which development does not proceed ($r_{\max}$ becomes zero). The combined impact of temperature and normality on the dissolution selectivity n is quite complicated.

Table 7.2 Developer temperature- and normality-dependent modeling parameters for Apex-E¹²

Developer Temperature (°C)	Developer Normality	$r_{\max}$ (nm/s)	$r_{\min}$ (nm/s)	m_{th}	n
20	0.195	45.9 ± 7.9	0.272	0.527 ± 0.03	5.26 ± 1.00
20	0.24425	90.9 ± 11.3	0.912	0.607 ± 0.018	5.41 ± 0.61
20	0.26	115.7 ± 14.1	1.482	0.682 ± 0.016	4.79 ± 0.49
35	0.195	78.1 ± 12.9	0.395	0.533 ± 0.01	17.33 ± 2.18
35	0.24425	151.3 ± 13.1	1.191	0.667 ± 0.008	11.10 ± 1.14
35	0.26	199.6 ± 14.6	2.562	0.693 ± 0.008	15.58 ± 2.17
40	0.195	118.5 ± 94.6	0.395	0.511 ± 0.074	8.72 ± 1.93
40	0.24425	182.9 ± 12.4	1.473	0.675 ± 0.005	16.44 ± 1.62
40	0.26	196.2 ± 13.1	2.135	0.694 ± 0.008	11.25 ± 1.32

One important reason for understanding developer normality effects is to better understand the impact of *developer loading*. One of the most popular styles of development is puddle development, where a puddle of developer is formed on top of the wafer which sits stationary through most of the development cycle. One potential issue with this development method, especially for thicker resists, is the accumulation of dissolved resist and the decrease in $-\text{OH}$ concentration in the developer as development proceeds in what is called developer loading. Additionally, if an integrated circuit has very different pattern densities across the die, this developer loading effect can vary across the die. Since development will slow down as the developer loading increases, the result can be an additional source of CD nonuniformity. A common solution to this problem is to break the puddle step into a *double puddle*, where each puddle is allowed to sit for half of the development time and fresh developer is used to form the second puddle.

7.2 The Development Contrast

The use of ‘contrast’ to describe the response of a photosensitive material dates back over 100 years. Hurter and Driffeld measured the optical density of photographic negative plates as a function of exposure.¹³ The ‘perfect negative’ was one that exhibited a linear variation of optical density with the logarithm of exposure. A plot of optical density versus log-exposure showed that a good negative exhibited a wide ‘period of correct representation’, as is shown in the *Hurter–Driffeld (H–D) curve* in Figure 7.10. Hurter and Driffeld called the slope of this curve in the linear region γ , the ‘development constant’. Negatives with high values of γ were said to be ‘high contrast’ negatives because the photosensitive emulsion quickly changed from low to high optical density when exposed. Of course, high contrast film is not always desirable in photography since it easily saturates.

7.2.1 Defining Photoresist Contrast

Photolithography evolved from the photographic sciences and borrowed many of its concepts and terminology. When exposing a photographic plate, the goal is to change the

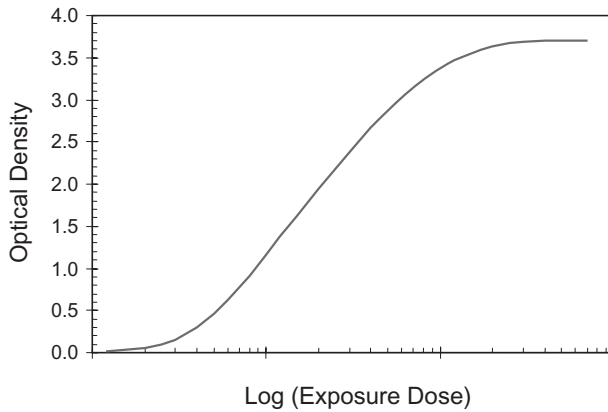


Figure 7.10 An example Hurter-Driffield (H-D) curve for a photographic negative

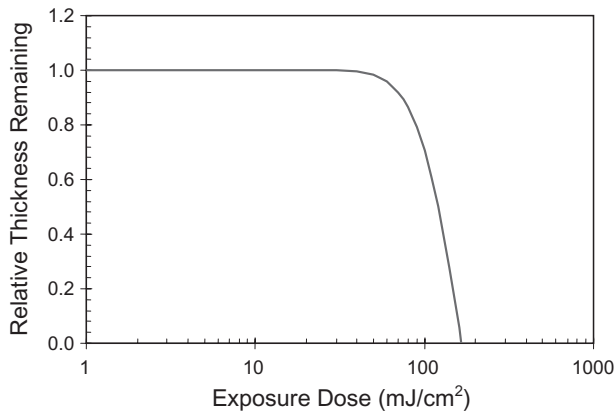


Figure 7.11 Positive resist characteristic curve used to measure contrast

optical density of the material. In lithography, the goal is to remove resist. Thus, one analogous H-D curve for lithography plots resist thickness after development versus log-exposure. Experimentally, a wafer is exposed using a blank glass plate as the mask. As each field on the wafer is exposed, the dose is adjusted so that a large array of doses is exposed on one wafer. After going through the normal PEB and development processes, the thickness remaining in this open-frame area is measured for each dose (Figure 7.11). The lithographic H-D curve (often called the *characteristic curve*) is a portion of a complete H-D curve as shown in Figure 7.10. Because the goal is to completely remove unwanted photoresist, there is usually a range of energies for which all of the photoresist is removed and thus the H-D curve would show no response. If, however, a very thick photoresist were used so that it could never be completely removed, the result would be the 'S'-shape of a complete H-D curve. It is common practice to normalize the initial

resist thickness to one, so that the lithographic H-D curve displays the relative thickness remaining.

Following the definition of γ from Hurter and Driffield, the *photoresist contrast* has traditionally been defined as the slope of the characteristic curve at the point where the thickness goes to zero. Thus,

$$\gamma_m = \pm \frac{1}{D} \frac{dz}{d \ln E} \bigg|_{E=E_0} \quad (7.21)$$

where z is the resist thickness removed during development, D is the resist thickness before development, E is the nominal exposure energy, and E_0 is the energy at which $z = D$. E_0 is called the clearing dose (or *dose-to-clear*) for positive photoresists and the gel dose for negative systems. The positive sign in Equation (7.21) is used for positive resists and the minus sign is used for negative systems in order to keep the value of γ_m positive. For the remainder of this section, positive systems will be used as our example. The results, however, can easily be applied to negative photoresists as well. [Note that sometimes a base-10 logarithm is used in Equation (7.21) rather than the natural logarithm. This book will always employ the natural log.]

Following the tradition of the photographic sciences, a high-contrast photoresist is one that makes a quick transition from being an ‘underexposed’ resist (which does not dissolve) to an ‘overexposed’ resist (which dissolves completely). The traditional definition in Equation (7.21) seems to fit this concept. Further, it is reasonably analogous to the photographic contrast as defined by Hurter and Driffield. The slope of optical density (which is a logarithm of transmittance) versus log-exposure is similar in form to the slope of relative thickness (which is like a log-thickness) versus log-exposure. Thus, it would seem that a suitable definition has been used.

However, there are numerous circumstances under which the definition in Equation (7.21) does not meet our expectations of what contrast means. In particular, the use of surfactant-laden developers dramatically increases measured contrast values, but without lithographic improvement. What causes these apparent discrepancies between the behavior of the measured contrast and our concept of how contrast affects lithography? Is there a problem with the definition, or with the measurement technique? These questions can be answered by putting the concept of contrast on a firm theoretical basis and applying rigorous analysis to the observed behavior.

With the advantage of retrospect, let us look at the evolution of the traditional definition of contrast and provide a different definition. In photography, the desired effect of exposure is a change in the optical density of the material. In photolithography, the desired effect is a change in development rate. This change in development rate is manifest as a change in resist thickness after development. Analogous to the photographic H-D curve, let us plot log-development rate versus log-exposure energy. Figure 7.12 shows the results for a typical photoresist. Note that this graph gives a complete H-D curve and does not cut off at some energy E_0 . We can give a new definition of contrast, called the *theoretical contrast*, as¹⁴

$$\gamma_{th} = \frac{d \ln r}{d \ln E} \quad (7.22)$$

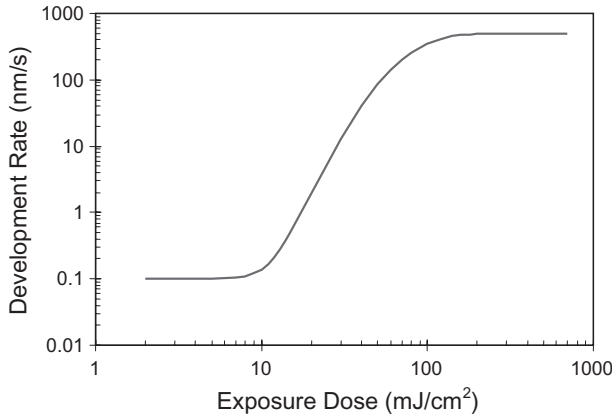


Figure 7.12 A lithographic H - D curve used to define the theoretical contrast

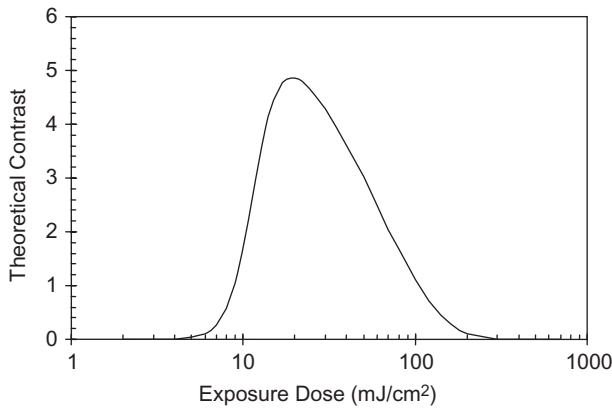


Figure 7.13 A typical variation of theoretical contrast with exposure dose. For this data, the $FWHM$ dose ratio is about 5.5

where r is the development rate. Note that the theoretical contrast will be a function of dose, going to zero at very low and very high doses. A typical plot of $\gamma_{th}(E)$ is shown in Figure 7.13. Not only is the maximum of this curve important, but the width of the $\gamma_{th}(E)$ function also affects lithographic performance. The equivalent of the full-width half-maximum ($FWHM$) on a logarithmic exposure scale would be the ratio of the high and low energies that give contrast equal to half of the maximum value, called the *FWHM dose ratio*.

The goal of lithographic exposure is to turn a gradient in exposure energy (an aerial image) into a gradient in development rate. From Equation 7.22, it is very easy to express this effect as

$$\frac{d \ln r}{dx} = \gamma_{th} \frac{d \ln I}{dx} \quad (7.23)$$

where I is the aerial image intensity and x is the horizontal distance from the center of the feature being printed. This equation is called the *Lithographic Imaging Equation* (and will be explored in more detail in Chapter 9). The left-hand term is the spatial gradient of development rate. To differentiate between exposed and unexposed areas, it is desirable to have this gradient as large as possible. The right-hand side of Equation (7.23) contains the log-slope of the aerial image. This term represents the quality of the aerial image, or alternatively the amount of information contained in the image about the position of the mask edge. In order to achieve a large development gradient between exposed and unexposed parts of the resist, one needs a large exposure gradient and a high photoresist contrast. Also, since contrast is a function of dose, it is important that the theoretical contrast stays high over a wide enough range of doses (that is, the FWHM dose ratio must be large). This is also equivalent to saying that $r_{\max}/r_{\min}$ must be large (see section 7.2.4).

7.2.2 Comparing Definitions of Contrast

The theoretical definition of contrast can now be compared to the conventional measured contrast, γ_m . The thickness remaining after development for the conventional H-D curve measurement can be described by integrating the definition of development rate for our open-frame exposures:

$$r = \frac{dz}{dt}, \quad \int_0^{t_{\text{dev}}} dt = t_{\text{dev}} = \int_0^z \frac{dz'}{r(z')} \quad (7.24)$$

where t_{dev} is the final development time. Knowing that development rate is a function of dose as well as depth into the resist, $r(E, z)$, and taking the derivative of both sides of the right-hand equation with respect to log-exposure,

$$0 = - \int_0^z \frac{dr}{d \ln E} \frac{dz'}{r^2(z')} + \frac{1}{r(z)} \frac{dz}{d \ln E} \quad (7.25)$$

Rearranging,

$$\frac{1}{D} \frac{dz}{d \ln E} = \frac{r(z)}{D} \int_0^z \frac{d \ln r}{d \ln E} \frac{dz'}{r(z')} \quad (7.26)$$

Evaluating this expression at $E = E_0$ (i.e. $z = D$) and using the definitions of measured and theoretical contrast,

$$\gamma_m = \frac{r(z=D)}{D} \int_0^D \gamma_{\text{th}} \frac{dz}{r(z)} \quad (7.27)$$

If the development rate does not vary through the resist thickness, it is easy to show that Equation (7.27) predicts that measured and theoretical contrast values are equal. But if the development rate varies with depth into the resist, the measured contrast fails to provide a good measurement of the theoretical contrast.

Consider the case of surface inhibition (caused, for example, by surfactants in the developer). The development rate at the top of the resist will be much lower than that at the bottom. Thus, Equation (7.27) shows that the measured contrast will be equal to the

theoretical contrast multiplied by some number bigger than 1. As the surface development rate goes to zero, the measured contrast becomes infinite, independent of the value of the theoretical contrast. Likewise, absorption makes the measured contrast value artificially low. Since every resist will exhibit some depth-dependent dissolution behavior, the conventional measured contrast is rarely an accurate measure of the true resist contrast. Both of these cases can be explored in detail using Equation (7.27).

For absorption, assume that the development rate follows the z -dependence given by Equation (7.19). Let's also assume that the theoretical contrast is constant (that is, through the thickness of the resist, all of the doses lie in the range of the approximately linear region of Figure 7.12). Thus,

$$\gamma_m = \frac{r(z=D)}{D} \int_0^D \gamma_{th} \frac{dz}{r(z)} = \frac{r(0)e^{-\alpha n D}}{D} \gamma_{th} \int_0^D \frac{dz}{r(0)e^{-\alpha n z}} \quad (7.28)$$

Solving the integral,

$$\gamma_m = \gamma_{th} \left(\frac{1 - e^{-\alpha n D}}{\alpha n D} \right) \quad (7.29)$$

The behavior of this result can be seen easily by considering the case of relatively small absorption ($\alpha n D \ll 1$) and taking a Taylor expansion of the exponential out to second order:

$$\gamma_m = \gamma_{th} \left(\frac{1 - e^{-\alpha n D}}{\alpha n D} \right) \approx \gamma_{th} \left(1 - \frac{\alpha n D}{2} \right) \quad (7.30)$$

Thus, absorption reduces the measured contrast by an amount approximately proportional to $\alpha n D$. Also, as will be seen in Chapter 9, the dissolution selectivity parameter n is directly proportional to the theoretical contrast. Thus, in the presence of absorption, any increase in the theoretical contrast will result in only a fractional increase in the measured contrast.

Surface inhibition will increase the measured contrast over the theoretical contrast. Consider again the case where theoretical contrast is constant, but let the depth dependence of the development rate be governed by the surface inhibition function of Equation (7.17). Further, assume that $D \gg \delta$ so that the development rate at the bottom of the resist is about equal to the bulk development rate. The measured contrast will become

$$\gamma_m = \frac{r_B}{D} \gamma_{th} \int_0^D \frac{dz}{r_B [1 - (1 - r_0)e^{-z/\delta}]} \quad (7.31)$$

Solving the integral,

$$\gamma_m = \gamma_{th} \left[1 - \frac{\delta}{D} \ln(r_0) \right] \quad (7.32)$$

As the relative surface rate r_0 goes to zero, the measured contrast goes to infinity. For example, if the surface inhibition depth is 20% of the resist thickness and the relative surface rate is 0.1, the measured contrast will be about 50% higher than the theoretical

contrast. If the relative surface rate is 0.01, the measured contrast will be about double the true value.

The problems with the characteristic curve as discussed above are often very severe. While absorption can be easily characterized for a given resist and the measured contrast corrected for absorption effects, surface inhibition is rarely characterized with sufficient accuracy to allow its impact to be removed from the measured contrast. As a result, few lithographers use the characteristic curve for anything but a dose-to-clear measurement. Instead, the theoretical contrast is most often measured directly using a development rate monitor (see section 7.4), if it is measured at all.

7.2.3 The Practical Contrast

One possible improvement in the measurement of photoresist contrast uses a method that Peter Gwozdz calls the ‘*practical contrast*’. Rather than measuring the resist thickness remaining as a function of dose for open-frame exposures, as for the conventional measurement based on the characteristic curve, the practical contrast measures the development time required to clear each open frame. For reasonably slowly developing resists (true for many g- and i-line resists), the clear time can be measured visually using a stopwatch. Plotting time-to-clear versus dose on a log-log scale gives a curve whose slope is equal to the negative of the theoretical contrast.

To see how this is true, consider a resist where the influence of dose on development rate is separable from the z -dependence.

$$r(E, z) = f(E)g(z) \quad (7.33)$$

Such separable behavior has been assumed for the previously discussed absorption and surface inhibition effects. The theoretical contrast can then be expressed as

$$\gamma = \frac{\partial \ln r}{\partial \ln E} = \frac{\partial \ln f}{\partial \ln E} \quad (7.34)$$

The time to clear will be given by Equation (7.24) when $z = D$:

$$t_{\text{clear}} = \int_0^D \frac{dz}{r(z)} = \frac{1}{f(E)} \int_0^D \frac{dz}{g(z)} \quad (7.35)$$

Since the integral on the right-hand side of Equation (7.35) is a constant (independent of dose), the time-to-clear will be inversely proportional to $f(E)$ and

$$\gamma = \frac{\partial \ln f}{\partial \ln E} = - \frac{\partial \ln t_{\text{clear}}}{\partial \ln E} \quad (7.36)$$

Unfortunately, accurate measurement of time-to-clear is often difficult without automated equipment such as a development rate monitor (see section 7.4). For g- and i-line resists, with maximum development rates of a few hundred nm/s and thicknesses greater than 500 nm, the smallest time-to-clear will be a few seconds. For many 248-nm chemically amplified resists, the maximum development rate is 1000–5000 nm/s and thicknesses are as low as a few hundred nanometers, so that over a fair range of doses the time-to-clear will be less than 1 second. Fortunately, though, the steep ‘high contrast’ portion of

the time-to-clear versus dose curve must necessarily (for any reasonably good resist) have time-to-clear values on the order of the development time (typically 45–90 seconds). Thus, if the goal is to measure the maximum contrast rather than the full contrast versus dose function, the practical contrast approach can work quite well.

7.2.4 Relationship Between γ and $r_{\max}/r_{\min}$

As was mentioned briefly at the end of section 7.1.1, a good development rate function not only has a high dissolution selectivity parameter n , but also a large ratio $r_{\max}/r_{\min}$. In the context of development contrast, these two requirements are equivalent to saying that the resist must have a high maximum contrast and a large FWHM dose ratio (see Figure 7.13). To more explicitly see the relationships at work, let's assume that our $\gamma(E)$ function is Gaussian in shape (on a log-exposure scale):

$$\gamma(E) = \gamma_{\max} e^{-(\ln E - \ln E^*)^2 / 2\sigma^2} \quad (7.37)$$

where $\gamma_{\max}$ is the maximum contrast, and E^* is the dose that gives the maximum contrast. The FWHM dose ratio will be given by

$$\frac{E_{1/2}}{E_{-1/2}} = \exp(2\sigma\sqrt{2\ln 2}) \approx \exp(2.355\sigma) \quad (7.38)$$

Integrating Equation (7.37) with respect to $\ln E$ gives an error function shape to the contrast curve. Using the limits of zero and infinite dose to define $r_{\max}$ and $r_{\min}$ gives

$$\ln\left(\frac{r}{\sqrt{r_{\max}r_{\min}}}\right) = \ln\left(\sqrt{\frac{r_{\max}}{r_{\min}}}\right) \operatorname{erf}\left(\frac{\ln(E/E^*)}{\sqrt{2}\sigma}\right) \quad (7.39)$$

where

$$\sigma = \frac{1}{\sqrt{2\pi}\gamma_{\max}} \ln\left(\frac{r_{\max}}{r_{\min}}\right) \quad (7.40)$$

Combining Equations (7.38) and (7.40),

$$\frac{1}{\gamma_{\max}} \ln\left(\frac{r_{\max}}{r_{\min}}\right) = \sqrt{\frac{\pi}{4\ln 2}} \ln\left(\frac{E_{1/2}}{E_{-1/2}}\right) \approx 1.06 \ln\left(\frac{E_{1/2}}{E_{-1/2}}\right) \quad (7.41)$$

The required FWHM dose ratio is determined by the aerial image and is proportional to the ratio of the maximum intensity in the space to the intensity at the line edge (which has typical values of 2–4). Thus, the required maximum to minimum development rate ratio goes as

$$\frac{r_{\max}}{r_{\min}} \sim e^{a\gamma_{\max}}, \quad 1 < a < 2 \quad \text{or} \quad \frac{r_{\max}}{r_{\min}} \sim \left(\frac{E_{1/2}}{E_{-1/2}}\right)^{\gamma_{\max}} \quad (7.42)$$

Equation (7.42) provides the interesting result that resists with higher contrast require a higher $r_{\max}/r_{\min}$ in order to make effective use of that contrast. A doubling of the contrast requires a squaring of the development rate ratio. For example, if the dose ratio is 2 and the maximum contrast is 10, the maximum to minimum development rate ratio should be

at least 1000. For the same dose ratio, however, a maximum contrast of 15 would require that $r_{\max}/r_{\min}$ be greater than 30 000 in order to take full advantage of the higher contrast.

7.3 The Development Path

Most of the discussion in this chapter so far has been concerned with how dose or resist chemical composition is converted to development rate. A spatial distribution of energy turns into a spatial distribution of chemicals which turns into a spatial distribution of dissolution rates. But how does a spatial distribution in development rates turn into an actual resist profile? How does the resist-developer front propagate through time and space during development?

For the one-dimensional case of large open-frame exposures (such as the characteristic curve experiment described in the previous section), propagation of the resist-developer interface is strictly in the z -direction, with the rate of development at a given point equal to the rate at which the interface moves down in z . For the more general case, we can define the path of development by tracing the position of the resist surface through the development time (Figure 7.14). This path will have a direction such that it is always perpendicular to the surface of the resist throughout the development cycle. Letting s be the length along this path at any given time, the development rate will then be

$$r = \frac{ds}{dt} \quad (7.43)$$

The problem then is to follow this path at a rate given by the spatially varying dissolution properties of the resist until the total development time is reached. Integrating this general definition of development rate thus gives a path integral:

$$t_{\text{dev}} = \oint_{\text{path}} \frac{ds}{r(x,y,z)} \quad (7.44)$$

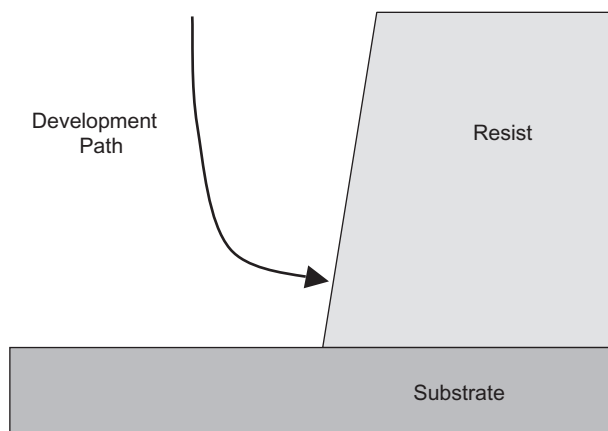


Figure 7.14 Typical development path

Before tackling the mathematical problem of solving this path integral, let's consider some of the general properties expected of this path. The path always begins vertically (since the resist surface is horizontal). If the final resist profile is close to our goal of nearly vertical sidewalls, the path must end going nearly horizontally. Thus, we should expect a path that starts out vertically, travels down through the resist thickness, then turns horizontally, stopping at the resist edge (see Figure 7.14).

7.3.1 The Euler–Lagrange Equation

The first step in solving the development path integral will be to parameterize the path in terms of its position. For simplicity, we will consider only the two-dimensional problem $s(x,z)$, though it can be easily extended to three dimensions. Consider a section of that path (Figure 7.15) small enough that it can be approximated as linear. It is clear that

$$\Delta s = \sqrt{\Delta x^2 + \Delta z^2} = \Delta x \sqrt{1 + \left(\frac{\Delta z}{\Delta x}\right)^2} \quad (7.45)$$

Letting the differentials go to zero,

$$ds = dx \sqrt{1 + z'^2} \quad (7.46)$$

where z' is the slope of the development path at any given point. The path integral then becomes

$$t_{\text{dev}} = \int_{x_0}^x \frac{dx \sqrt{1 + z'^2}}{r(x,z)} \quad (7.47)$$

where the path starts at $(x_0, 0)$ and ends at (x, z) .

Obviously, to solve Equation (7.47), one must know not only the development rate as a function of position, but the slope of the path at every point along the path as well. But how is one to determine the path of development? Physical insight and mathematical methods must both be exploited. The physical constraint on what path is chosen is called the *principle of least action*: the path taken will be that which goes from the starting

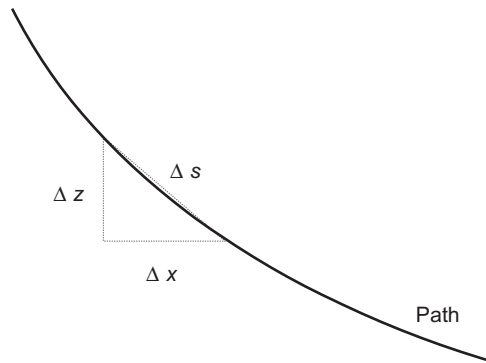


Figure 7.15 Section of a development path relating path length s to x and z

point to the finish point in the least amount of time. To see why this must be true, just imagine a different path competing with the least-time path. By the time the competing path made it to the point (x,z) , the faster path would have already reached there, and thus would have removed the photoresist from that point! Thus, only the least-time path survives.

The principle of least action means that a path must be found that minimizes the integral in Equation (7.47). A fundamental result from the calculus of variations states that the general integral

$$\int f(x,z,z')dx \quad (7.48)$$

will have a minimum or maximum whenever the Euler–Lagrange equation

$$\frac{\partial f}{\partial z} - \frac{d}{dx} \left(\frac{\partial f}{\partial z'} \right) = 0 \quad (7.49)$$

is satisfied. Applying the Euler–Lagrange approach to the development path, the least-time path will be that which satisfies the second-order nonlinear differential equation:

$$z'' + (1 + z'^2) \left(\frac{\partial \ln r}{\partial z} - z' \frac{\partial \ln r}{\partial x} \right) = 0 \quad (7.50)$$

(As an interesting aside, there are many problems in physics where the principle of least action is coupled with the Euler–Lagrange technique to solve problems. In optics, techniques in geometrical optics are used to trace rays of light through materials of different refractive indices. Fermat’s principle says light will use the path that minimizes the travel time – a principle of least action. The rays follow paths equivalent to the development path discussed here. The optical path length, or equivalently the optical phase, represents the development time in this problem, and a wavefront – a surface of constant phase – is equivalent to the resist profile – a surface of constant development time. The refractive index is equivalent to $1/r$.)

An equivalent formulation of the Euler–Lagrange equation is the Hamilton–Jacobi equation. A surface of constant development time (i.e. the resist surface) is defined by the Hamilton–Jacobi equation:

$$\left(\frac{\partial t}{\partial x} \right)^2 + \left(\frac{\partial t}{\partial z} \right)^2 = \frac{1}{r^2} \quad (7.51)$$

(This equation is equivalent to the Eikonal equation of geometrical optics.) The Hamilton–Jacobi formulation is useful for ‘ray-trace’-like calculations, where at each point the dissolution rate is broken down into x - and z -components and the trajectory of the ray is traced out over time.

7.3.2 The Case of No z -Dependence

Solutions to Equation (7.50) for general development problems will almost always require extensive numerical techniques. However, there are a few simplifying cases that yield analytical solutions. Consider the idealized case where the development rate does

not vary with z (no absorption, no standing waves, etc.). The Euler–Lagrange equation simplifies to

$$z'' = z'(1 + z'^2) \frac{\partial \ln r}{\partial x} \quad (7.52)$$

This equation can be solved by direct integration with the boundary condition that $z' = \infty$ and $x = x_0$ at $z = 0$ (i.e. the path starts out vertically at $x = x_0$), giving

$$\frac{r(x)}{r(x_0)} = \frac{z'}{\sqrt{1 + z'^2}} \quad \text{or} \quad z' = \frac{r(x)}{\sqrt{r^2(x_0) - r^2(x)}} \quad (7.53)$$

The actual path $z(x)$ can be derived by integrating Equation (7.53),

$$z = \int_{x_0}^x \frac{r(x)}{\sqrt{r^2(x_0) - r^2(x)}} dx \quad (7.54)$$

and the development time becomes, by substituting Equation (7.53) into Equation (7.47),

$$t_{\text{dev}} = \int_{x_0}^x \frac{r(x_0)}{r(x) \sqrt{r^2(x_0) - r^2(x)}} dx \quad (7.55)$$

where the point (x, z) is the end of the path and thus a point on the final resist profile. For a given spatial distribution of development rates $r(x)$, the final resist profile is determined by solving the integrals (7.54) and (7.55) for many different values of x_0 (i.e. for different starting points for the paths), as shown in Figure 7.16, and connecting the paths to form a surface of constant development time.

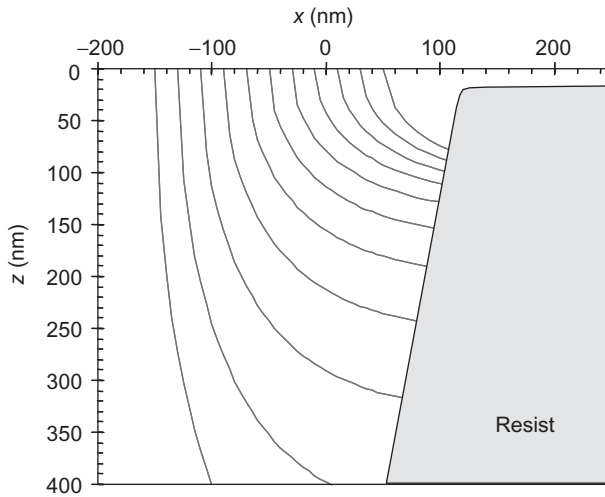


Figure 7.16 Example of how the calculation of many development paths leads to the determination of the final resist profile

An interesting form of Equation (7.55) reveals some physical insight into the problem. Combining Equations (7.55) and (7.53) in a different way,

$$t_{\text{dev}} = \frac{1}{r(x_0)} \int_{x_0}^x \frac{1+z'^2}{z'} dx = \frac{1}{r(x_0)} \int_{x_0}^x \frac{1}{z'} dx + \frac{z(x)}{r(x_0)} \quad (7.56)$$

The right-hand side of this equation seems to segment the development into horizontal and vertical components. The term $z/r(x_0)$ is the time required to develop vertically, starting at x_0 , to the value of z equal to the bottom of the path. The first term on the right-hand side (the integral) is then the effective amount of time spent developing horizontally from x_0 to x .

7.3.3 The Case of a Separable Development Rate Function

Another interesting case to examine is when the development rate function $r(x,z)$ is separable into a function of incident dose E only and a function of the spatial variables independent of dose:

$$r(E,x,z) = f(E)g(x,z) \quad (7.57)$$

In general, none of the kinetic development models discussed earlier meet this criterion. However, such separability is approximately true for the reaction-controlled original kinetic development model for conventional resists. For moderately low doses and no bleaching, the latent image is approximately

$$m(x,z) = \exp[-CEI_r(x,z)] \approx 1 - CEI_r(x,z) \quad (7.58)$$

Ignoring the minimum development rate, this approximate latent image gives a development rate proportional to a power of dose:

$$r(x,z) = r_{\text{max}}(1 - m(x,z))^n + r_{\text{min}} \approx r_{\text{max}}(CE)^n (I_r(x,z))^n \quad (7.59)$$

Obviously, the restrictions and approximations placed on the exposure and development functions to achieve separability are numerous and severe. Nonetheless, the separability assumption can be very useful. Looking at it another way, separability can be achieved by assuming a constant theoretical contrast. Integrating the definition of theoretical contrast, Equation (7.22), for the case of constant contrast, gives

$$r(x,z) \propto E^\gamma (I_r(x,z))^\gamma \quad (7.60)$$

Equations (7.59) and (7.60) are equivalent if $n = \gamma$. Thus, separability can be assumed if the range of dose experienced by the resist over the development path is sufficiently small to fall within the approximately linear region of the H-D curve.

Separability leads to an interesting result. Using Equation (7.57),

$$\frac{\partial \ln r}{\partial z} = \frac{\partial \ln g}{\partial z}, \quad \frac{\partial \ln r}{\partial x} = \frac{\partial \ln g}{\partial x} \quad (7.61)$$

and the Euler-Lagrange Equation (7.50) is completely independent of incident dose. In other words, as the dose is changed, the path of development does not change – the

development just follows that path at a different rate. Applying this idea to the defining development path Equation (7.47),

$$f(E)t_{\text{dev}} = \int_{x_0}^x \frac{dx\sqrt{1+z'^2}}{g(x,z)} = c, \quad \text{a constant} \quad (7.62)$$

For example, let $f(E) = E^\gamma$. If the dose is increased, the same path is used to get from x_0 to x , but it arrives in a shorter time.

7.3.4 Resist Sidewall Angle

The development path can help to understand even more about the resist profile. Since the path is always perpendicular to the resist profile, knowing the path gives us the sidewall angle of the profile (the angle that the profile makes with the horizontal substrate) – the tangent of the sidewall angle will be one over the development path slope. For the case of no z -variation in development rate, Equation (7.53) leads to

$$\cos \theta = \frac{r(x)}{r(x_0)} \quad (7.63)$$

The goal is to make the sidewall angle close to 90° , and thus make $\cos \theta \approx 0$. This is accomplished by making the development rate at the end of the path very small compared to the rate at the beginning of the path. In other words, good resist profiles are obtained when very little time is spent developing vertically compared to the time spent developing horizontally.

As previously mentioned, numerical methods are required for a general solution for the development path. However, an interesting limit can be observed that sheds further light on the final resist sidewall angle. For the case of no variation in development rate with z , the sidewall angle is given by Equation (7.63) and can be made as close to 90° as possible by making the development rate at the end of the path very small. However, in the presence of a depth dependence to the development rate, there is a different limiting sidewall angle. Consider the Euler–Lagrange Equation (7.50) near the end of the path assuming a reasonably good resist profile. For such a case, the path slope will be small and $z'^2 \ll 1$, so that

$$z'' \approx z' \frac{\partial \ln r}{\partial x} - \frac{\partial \ln r}{\partial z} \quad (7.64)$$

The minimum possible value of z' (and thus the best possible resist sidewall angle) will occur when $z'' = 0$. Thus,

$$z'_{\min} = \frac{\partial \ln r / \partial z}{\partial \ln r / \partial x} \quad (7.65)$$

Consider again the case of absorption causing the only variation in development rate with depth into the resist. Using our simple absorption model in Equation (7.19), the

Lithographic Imaging Equation (7.23), and approximating the dissolution selectivity parameter with the contrast,

$$z'_{\min} \approx \frac{\alpha}{\partial \ln I / \partial x} \quad (7.66)$$

The minimum path slope with absorption given in Equation (7.66) is obtained when the development rate at the end of the path goes to zero. For the case of finite development rate, the path slope near the end of path is given approximately by

$$z'_{\text{end}} \approx \frac{\alpha}{\partial \ln I / \partial x} + \frac{r(x)}{r(x_0)} \quad (7.67)$$

This result will be derived in the next section for a specific case, but is approximately true in general.

7.3.5 The Case of Constant Development Gradients

Another interesting case that can be explored analytically has constant spatial log-gradients of the development rate. Let

$$\frac{\partial \ln r}{\partial x} = -a, \quad \frac{\partial \ln r}{\partial z} = -b \quad (7.68)$$

where a and b are constants. Interpreting these constants as before, a is something like the resist contrast times image log-slope. The term b is the contrast multiplied by the absorption coefficient. Note that both a and b will be positive. The Euler–Lagrange equation can be solved by direct integration, with the boundary condition that $z' = \infty$ and $x = x_0$ at $z = 0$. After some algebraic manipulation of the result,

$$\exp \left[-a(x - x_0) \left(1 + \frac{b^2}{a^2} \right) \right] = \left[\frac{z' - b/a}{\sqrt{1 + z'^2}} \right] \exp \left[\frac{b}{a} \cot^{-1}(z') \right] \quad (7.69)$$

For a typical process, b/a is in the range of 0.02–0.05 (development gradients in x must be significantly greater than the gradients in z if a good resist profile is to result). Thus, it is reasonable to neglect b^2/a^2 compared to 1. Also, the assumption of constant log-gradient of development rates means that the development rate is separable into x - and z -components and

$$e^{-a(x-x_0)} = \frac{r(x)}{r(x_0)} \quad (7.70)$$

Thus, Equation (7.69) becomes

$$\frac{r(x)}{r(x_0)} \approx \left[\frac{z' - b/a}{\sqrt{1 + z'^2}} \right] \exp \left[\frac{b}{a} \cot^{-1}(z') \right] \quad (7.71)$$

Finally, the inverse cotangent will be zero at the beginning of the path (when $z' = \infty$), going to $\pi/4$ for $z' = 1$, and reaching its maximum of $\pi/2$ as z' goes to zero. Given the

small value of b/a , the exponential term can be approximated with a Taylor's series to give

$$\frac{r(x)}{r(x_0)} \approx \left(1 + \frac{b}{a} \cot^{-1}(z')\right) \left(\frac{z' - b/a}{\sqrt{1 + z'^2}}\right) \quad (7.72)$$

For a small enough value of b/a ,

$$\frac{r(x)}{r(x_0)} \approx \left(\frac{z' - b/a}{\sqrt{1 + z'^2}}\right) \quad (7.73)$$

Note that at the end of the path, $z'^2 \ll 1$, and the result given previously in Equation (7.67) is obtained.

Before continuing with the solution, let's consider a simpler case where $b = 0$. Thus, we are solving the no z -dependence case with the development rate function given in Equation (7.70). Substituting this rate expression into Equation (7.54) and integrating gives us the path $z(x)$ for a given starting point x_0 :

$$z(x, x_0) = \frac{1}{a} \tan^{-1}(\sqrt{e^{2a(x-x_0)} - 1}) = \frac{1}{a} \cos^{-1}(e^{-a(x-x_0)}) \quad (7.74)$$

Going on to find the development time at each point on the path by carrying out the integration in Equation (7.55),

$$t_{\text{dev}} = \frac{1}{ar(x_0)} \sqrt{e^{2a(x-x_0)} - 1} \quad (7.75)$$

The resist profile sidewall angle is simply az .

A convenient way to express this result is to parameterize positions along the path (and thus on the resist profile) as a function of the path starting point and the development time. Rearranging the results in Equations (7.74) and (7.75),

$$z(x_0, t) = \frac{1}{a} \tan^{-1}(ar(x_0)t), \quad x(x_0, t) = x_0 + \frac{1}{a} \ln(\sqrt{[ar(x_0)t]^2 + 1}) \quad (7.76)$$

Note that for the very likely case that $ar(x_0)t \gg 1$,

$$x(x_0, t) = x_0 + \frac{1}{a} \ln(ar(x_0)t) \quad (7.77)$$

When absorption is added, the results are in general complicated, but simplify when one doesn't care to know the resist profile at small development times. For normal conditions, the path and develop time can be expressed as the no-absorption solution plus an added term:

$$\begin{aligned} z(x, x_0) &= \frac{1}{a} \tan^{-1}(\sqrt{e^{2a(x-x_0)} - 1}) + \frac{b}{a}(x - x_0) \\ t_{\text{dev}} &= \frac{1}{ar(x_0)} \sqrt{e^{2a(x-x_0)} - 1} + \frac{b}{ar(x_0)}(x - x_0) \end{aligned} \quad (7.78)$$

Thus, absorption will slow down the development, making the time required to get to a certain value of x longer. Also, since the path never levels off to be completely horizontal, continued development pushes the profile down (larger z) as well as to the right.

7.3.6 Segmented Development and the Lumped Parameter Model (LPM)

Consider again the case of a constant theoretical contrast. Equation (7.60), for the simple 2D case, can be expressed as

$$r(x, z) = r_0 \left(\frac{E}{E_0} \right)^\gamma (I(x)I_z(z))^\gamma \quad (7.79)$$

where the intensity variation is assumed separable into x - and z -components and E_0 is any arbitrary reference dose that produces the development rate r_0 for an open-frame exposure (so that $I(x)I_z(0) = 1$). It will be convenient to let this dose be the dose-to-clear. Applying Equation (7.24) at this dose,

$$t_{\text{dev}} = \int_0^D \frac{dz}{r(z)} = \frac{1}{r_0} \int_0^D \frac{dz}{(I_r(z))^\gamma} \quad (7.80)$$

An effective resist thickness can be defined as

$$D_{\text{eff}} = r(E_0, z = D)t_{\text{dev}} = \int_0^D \left(\frac{I_z(D)}{I_z(z)} \right)^\gamma dz \quad (7.81)$$

It is clear that for no variation in intensity with depth into the resist this effective resist thickness will equal the actual resist thickness.

As an example, the effective resist thickness can be calculated for the case when only absorption causes a variation in intensity with depth into the resist. For such a case, $I(z)$ will decay exponentially with an absorption coefficient α , and Equation (7.81) can be evaluated to give

$$D_{\text{eff}} = \frac{1}{\alpha\gamma} (1 - e^{-\alpha\gamma D}) \quad (7.82)$$

Equation (7.79) is an extremely simple-minded model relating development rate to exposure energy based on the assumption of a constant resist contrast. In order to use this expression, we still need to determine the path of dissolution. However, rather than calculate the path for some given development variation $r(x, z)$, it will be useful to assume a reasonable development path and then use this path to calculate the final resist CD as a function of exposure dose. The assumed path will be the *segmented development path*:¹⁵ the path goes vertically to a depth z , followed by a lateral development to position x .

A development ray, which traces out the path of development, starts at the point $(x_0, 0)$ and proceeds vertically until a depth z is reached, at which point the development will begin horizontally. Since our interest is to determine the bottom linewidth, consider a path

where the depth z is just equal to the resist thickness D . The vertical development time becomes

$$t_z = \frac{1}{r(x_0, D)} \int_0^D \left(\frac{I(z)}{I(D)} \right)^{-\gamma} dz = \frac{D_{\text{eff}}}{r(x_0, D)} \quad (7.83)$$

Similarly, the horizontal development time becomes

$$t_x = \frac{1}{r(x_0, D)} \int_{x_0}^x \left(\frac{I(x')}{I(x_0)} \right)^{-\gamma} dx' \quad (7.84)$$

The total development time is still the sum of these two components.

$$t_{\text{dev}} = \frac{D_{\text{eff}}}{r(x_0, D)} \left(1 + \frac{1}{D_{\text{eff}}} \int_{x_0}^x \left(\frac{I(x')}{I(x_0)} \right)^{-\gamma} dx' \right) \quad (7.85)$$

By applying Equations (7.79) and (7.81),

$$\frac{r(x_0, D)t_{\text{dev}}}{D_{\text{eff}}} = \left(\frac{E(x)}{E(0)} \right)^{\gamma} \quad (7.86)$$

where $E(x)$ is the dose required to create a $\text{CD} = 2x$, and $E(0)$ is the dose required to make the CD just equal to zero. Note that $E(0)$ is not the dose-to-clear since in general $I(x_0)$ will not be equal to 1.

Substituting Equation (7.86) into (7.85) gives the Lumped Parameter Model.¹⁶

$$\left(\frac{E(x)}{E(0)} \right)^{\gamma} = 1 + \frac{1}{D_{\text{eff}}} \int_{x_0}^x \left(\frac{I(x')}{I(x_0)} \right)^{-\gamma} dx' \quad (7.87)$$

Given any aerial image $I(x)$, a CD -versus-dose curve can be calculated by evaluating this integral numerically. For a small space (or small dense lines and spaces), the start of the path x_0 will always be the center of the space. The only parameters that control the relationship between the shape of the aerial image and the shape of the exposure latitude curve are the resist contrast and the effective resist thickness. As the effective resist thickness goes to zero and the resist contrast goes to infinity, the dose required to produce a CD of $2x$ becomes proportional to $1/I(x)$.

7.3.7 LPM Example – Gaussian Image

In the clear region of an image, near the edge of a line, the aerial image (and the image in resist) can quite often be reasonably approximated as a Gaussian function:

$$I(x) = I_0 e^{-x^2/2\sigma^2} \quad (7.88)$$

Since only the space region develops away in a positive resist to form the resist profile, an accurate representation of the image is needed only in this region that develops away. The reason why an aerial image of a space often looks Gaussian in shape can be seen by looking at Taylor expansions of the Gaussian and of a typical cosine Fourier

representation of an image. Consider for example an image with two Fourier components (a three-beam imaging case) expanded in a Taylor series about $x = 0$ (the center of the space) for a given pitch p :

$$\begin{aligned} I(x) &= a_0 + a_1 \cos(2\pi x/p) + a_2 \cos(4\pi x/p) \\ &= (a_0 + a_1 + a_2) - \left(\frac{a_1 + 4a_2}{2}\right)(2\pi x/p)^2 + \left(\frac{a_1 + 16a_2}{24}\right)(2\pi x/p)^4 - \dots \end{aligned} \quad (7.89)$$

Likewise for the Gaussian image:

$$I(x) = I_0 e^{-x^2/2\sigma^2} = I_0 - \left(\frac{I_0}{2\sigma^2}\right)x^2 + \left(\frac{I_0}{8\sigma^4}\right)x^4 - \dots \quad (7.90)$$

The zero- and second-order terms of these two series can be made to match exactly when

$$I_0 = a_0 + a_1 + a_2 \quad \text{and} \quad \sigma^2 = \left(\frac{a_0 + a_1 + a_2}{a_1 + 4a_2}\right) \left(\frac{p}{2\pi}\right)^2 \quad (7.91)$$

In addition, the fourth-order terms for each series are often quite close to each other in magnitude. For example, if $a_0 = a_1$ and $a_2 = 0.2a_0$ (a reasonable set of values), then a Gaussian image with parameters determined by Equation (7.91) will have fourth-order terms in the Taylor series for the aerial image and the Gaussian representation that differ in magnitude by only 5%. Further, by adjusting σ of the Gaussian slightly from the value given by Equation (7.91), a better overall match to the image in the region of the space can be obtained (see Figure 7.17).

The log-slope of the Gaussian image is

$$ILS = \frac{d \ln I}{dx} = \frac{x}{\sigma^2} \quad (7.92)$$

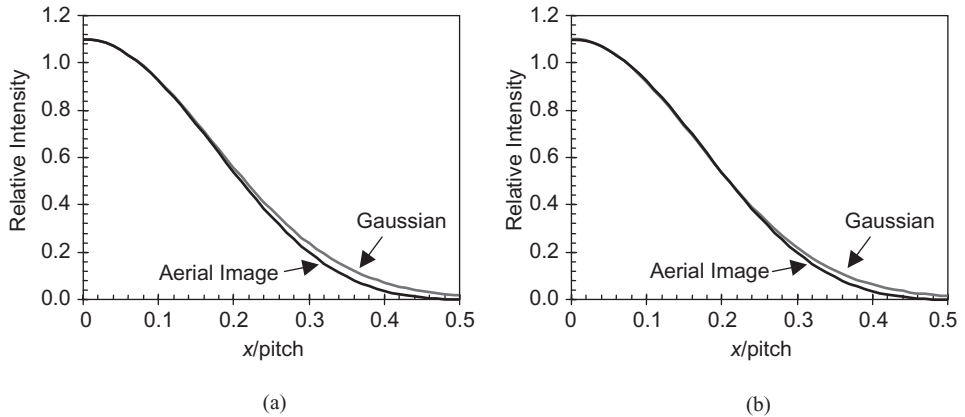


Figure 7.17 Matching a three-beam image ($a_0 = 0.45$, $a_1 = 0.55$, $a_2 = 0.1$) with a Gaussian using (a) a value of σ given by Equation (7.91); and (b) the best-fit σ . Note the goodness of the match in the region of the space ($x/\text{pitch} < 0.25$)

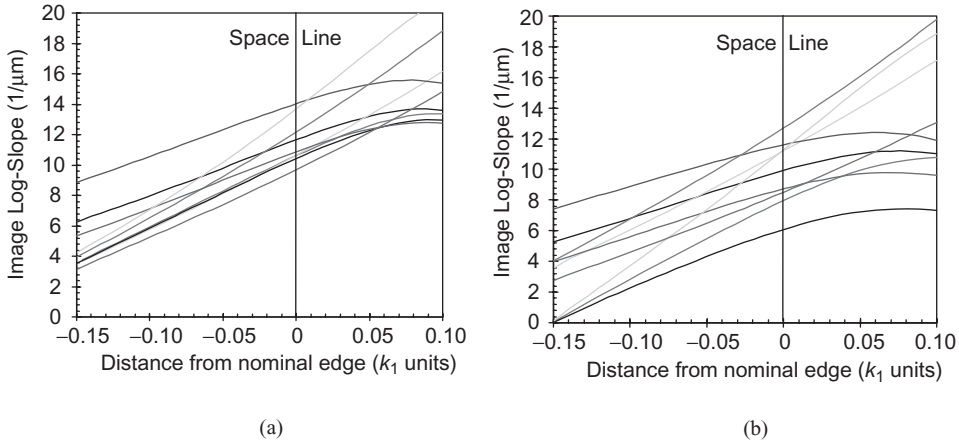


Figure 7.18 Simulated aerial images over a range of conditions show that the image log-slope varies about linearly with distance from the nominal resist edge in the region of the space: (a) $k_1 = 0.46$ and (b) $k_1 = 0.38$ for isolated lines, isolated spaces, and equal lines and spaces and for conventional, annular and quadrupole illumination

Thus, the log-slope varies linearly with x , a result that is approximately true for most images in the region of the space (Figure 7.18). If the nominal CD is CD_1 , then the normalized image log-slope (NILS) at this point is

$$NILS_1 = CD_1 \left. \frac{d \ln I}{dx} \right|_{x=CD_1/2} = \frac{CD_1^2}{2\sigma^2} \quad (7.93)$$

Putting the Gaussian aerial image into these terms,

$$I(x) = I_0 e^{-NILS_1(x/CD_1)^2} \quad (7.94)$$

Using this Gaussian image in the LPM Equation (7.87) and letting the development path start at $x = 0$,

$$\left(\frac{E(x)}{E(0)} \right)^\gamma = 1 + \frac{1}{D_{\text{eff}}} \int_0^x e^{\gamma NILS_1(x'/CD_1)^2} dx' \quad (7.95)$$

The integral is solvable in terms of the imaginary error function $erfi$. Letting $g = \gamma NILS_1/CD_1^2$,

$$\left(\frac{E(x)}{E(0)} \right)^\gamma = 1 + \frac{1}{2D_{\text{eff}}} \sqrt{\frac{\pi}{g}} erfi(\sqrt{g}x) \quad (7.96)$$

Since $erfi$ goes to infinity as x goes to infinity, a more convenient form can be obtained by relating the imaginary error function to the Dawson's integral $D_w(z)$:

$$erfi(z) = \frac{2}{\sqrt{\pi}} e^{z^2} D_w(z) \quad \text{where} \quad D_w(z) = e^{-z^2} \int_0^z e^{x^2} dx \quad (7.97)$$

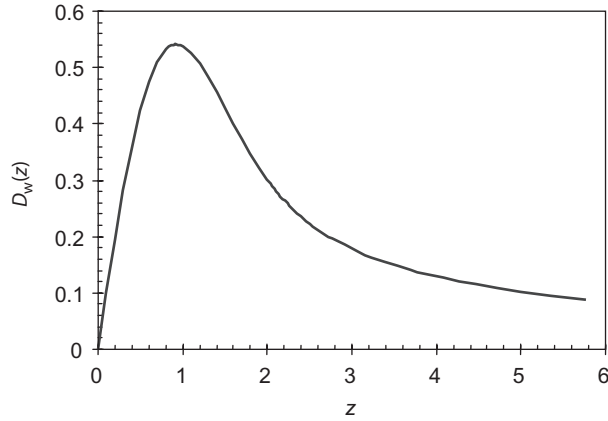


Figure 7.19 A plot of Dawson's integral, $D_w(z)$

Dawson's integral is a bit easier to handle since its maximum value is about 0.54 at $z \approx 0.92$ and goes to zero at large z (see Figure 7.19). The LPM result becomes

$$\begin{aligned} \left(\frac{E(x)}{E(0)} \right)^\gamma &= 1 + \frac{1}{D_{\text{eff}} \sqrt{g}} e^{gx^2} D_w(\sqrt{g}x) \\ &= 1 + \frac{1}{D_{\text{eff}} \sqrt{g}} \left(\frac{I(x)}{I(0)} \right)^{-\gamma} D_w(\sqrt{g}x) \end{aligned} \quad (7.98)$$

For large arguments, the Dawson's integral can be approximated reasonably well as

$$D_w(z) = \frac{1}{2z} + \frac{1}{4z^3} + \dots \approx \frac{1}{2z} \quad (7.99)$$

This final approximation has about 20% error when $z = 2$, 11% error at $z = 2.5$, 7% error at $z = 3$, and 3.5% error at $z = 4$. Small arguments occur for low g (low contrast resists or low NILS values) and low x (underexposed spaces where the CD is much lower than the nominal). This is also the regime where e^{gx^2} is small so that the impact of this error is much reduced. For typical cases, and where the CD being predicted by Equation (7.98) is within 50% of the nominal CD, the error in calculating the dose required to achieve that CD caused by using the approximation for the Dawson's integral is usually less than 1–2%.

In this large argument case, the resulting exposure response expression becomes

$$\left(\frac{E(x)}{E(0)} \right)^\gamma \approx 1 + \frac{1}{D_{\text{eff}} g(2x)} \left(\frac{I(x)}{I(0)} \right)^{-\gamma} \quad (7.100)$$

Substituting the definition of g and letting $CD = 2x$,

$$\left(\frac{E(x)}{E(0)} \right)^\gamma \approx 1 + \frac{1}{\gamma} \left(\frac{1}{D_{\text{eff}}} \right) \left(\frac{CD_1^2}{CD \text{NILS}_1} \right) \left(\frac{I(x)}{I(0)} \right)^{-\gamma} \quad (7.101)$$

For our Gaussian image [and from Equations (7.92) and (7.93)],

$$ILS = \frac{CD}{2\sigma^2} = \frac{CDNILS_1}{CD_1^2} \quad (7.102)$$

which gives the final expression:

$$\frac{E(x)}{E(0)} \approx \left[1 + \left(\frac{1}{\gamma D_{\text{eff}} ILS(x)} \right) \left(\frac{I(x)}{I(0)} \right)^{-\gamma} \right]^{\frac{1}{\gamma}} \quad (7.103)$$

Remembering that log-slope of the aerial image is also a function of x , Equation (7.103) predicts the dose required to achieve a certain CD ($= 2x$) as a function of the aerial image $I(x)$, the resist contrast γ , and the effective thickness of the resist D_{eff} .

Equation (7.103) can be further simplified by assuming that the vertical development time of the LPM can be ignored compared to the horizontal development time (an assumption that is reasonable for a good lithography process, but may not be accurate when trying to understand features that are just able to print, for example at the extremes of focus). In this case, the 1 in the square brackets of Equation (7.103) can be ignored, giving

$$E(x) \approx \frac{E(0)I(0)}{I(x)} \left(\frac{1}{\gamma D_{\text{eff}} ILS} \right)^{\frac{1}{\gamma}} = \frac{E_0}{I(x)} \left(\frac{1}{\gamma D_{\text{eff}} ILS} \right)^{\frac{1}{\gamma}} \quad (7.104)$$

This interesting result is in fact an example of a threshold resist model: the resist CD is obtained when the local dose $EI(x)$ reaches a certain threshold dose E_{th} .

$$E = \frac{E_{\text{th}}}{I(x)} \quad \text{where} \quad E_{\text{th}} = E_0 \left(\frac{1}{\gamma D_{\text{eff}} ILS} \right)^{\frac{1}{\gamma}} \quad (7.105)$$

The particular variant of the threshold resist model derived here is called a variable-threshold resist model (sometimes called a VTR model). The threshold varies as a function of the image log-slope of the image. This model also relates the threshold to physically significant resist parameters: E_0 , γ and D_{eff} .

Figure 7.20 compares predictions of the CD of a space versus exposure dose for three different models: the LPM Equation (7.98), the LPM using the approximation for the Dawson's integral of Equation (7.103), and the VTR of Equation (7.105). For the conditions here, the approximate form of Dawson's integral is quite accurate and the approximate LPM expression is hard to distinguish from the full LPM results. As expected, the VTR begins to deviate from the LPM at lower doses, where the vertical development time is a significant fraction of the total development time. At the nominal CD, the approximate LPM predicts the dose wrong by 1%, and the VTR is off by 2%. For a CD that is 10 and 20% too small (underexposed), the approximate LPM is still off by only 1% for both, whereas the VTR misses by about 3.5 and 6.5%, respectively. Overexposed (CD of the space getting larger), the VTR and approximate LPM both improve in accuracy.

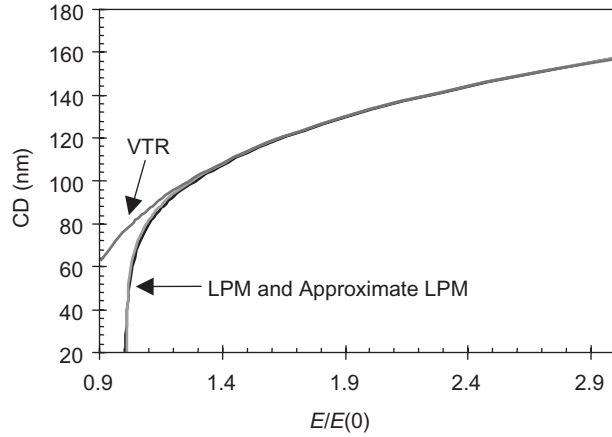


Figure 7.20 CD-versus-dose curves as predicted by the LPM Equation (7.98), the LPM using the approximation for the Dawson's integral, and the VTR. All models assumed a Gaussian image with $g = 0.0025 \text{ 1/nm}^2$ ($NILS_1 = 2.5$ and $CD_1 = 100 \text{ nm}$), $\gamma = 10$, and $D_{\text{eff}} = 200 \text{ nm}$

An alternate (and usually more convenient) form of this VTR model can be obtained by inserting the Gaussian image intensity [Equation (7.94)] that was used to derive the VTR model directly into Equation (7.104). After some algebraic manipulations, and letting E_1 be the dose that produces the reference spacewidth CD_1 ,

$$\frac{CD}{CD_1} = \sqrt{1 + \frac{4}{NILS_1} \left[\ln\left(\frac{E}{E_1}\right) + \frac{1}{\gamma} \ln\left(\frac{CD}{CD_1}\right) \right]} \quad (7.106)$$

Unfortunately, the equation is recursive with respect to CD . However, for dimensions close to CD_1 , the following approximation can be made:

$$\frac{\ln\left(\frac{CD}{CD_1}\right)}{\ln\left(\frac{E}{E_1}\right)} = \frac{\ln\left(1 + \frac{\Delta CD}{CD_1}\right)}{\ln\left(1 + \frac{\Delta E}{E_1}\right)} \approx \frac{\frac{\Delta CD}{CD_1}}{\frac{\Delta E}{E_1}} \approx \frac{2}{NILS_1} \quad (7.107)$$

Thus,

$$\frac{CD}{CD_1} \approx \sqrt{1 + \frac{4}{NILS_1} \ln\left(\frac{E}{E_1}\right) \left[1 + \frac{2}{\gamma NILS_1} \right]} \quad (7.108)$$

7.4 Measuring Development Rates

In general, development model parameters for a given resist/develop process are obtained by making real-time *in-situ* measurements of resist thickness during development using a development rate monitor (DRM). The resist thickness-versus-develop time curves are measured at many different incident exposure doses. Taking the derivative of this data gives development rate (r) as a function of incident dose (E) and depth into the resist (z).

By modeling the exposure process, a given exposure dose turns into a distribution of inhibitor concentration (m) after exposure as a function of depth. Combining the measured $r(E,z)$ with the calculated $m(E,z)$ produces a resultant $r(m,z)$ data set. These data are then fit with a development model (possibly including a surface inhibition function) and parameters for the model are extracted.

Although the best method for obtaining new parameters is the use of a DRM as described above, there is another approximate approach called the Poor Man's DRM.¹⁷ The Poor Man's DRM involves the measurement of multiple contrast curves – resist thickness remaining as a function of exposure dose for open-frame exposures – at different development times (Figure 7.21).

Analyzing the information in Figure 7.21 to obtain development rate, and eventually development parameters, is not trivial. The thickness as a function of dose and development time is first converted to rate as a function of dose and depth into the resist, $r(E,z)$. From here, analysis is the same as for conventional DRM data. Figure 7.22 shows an example set of results.

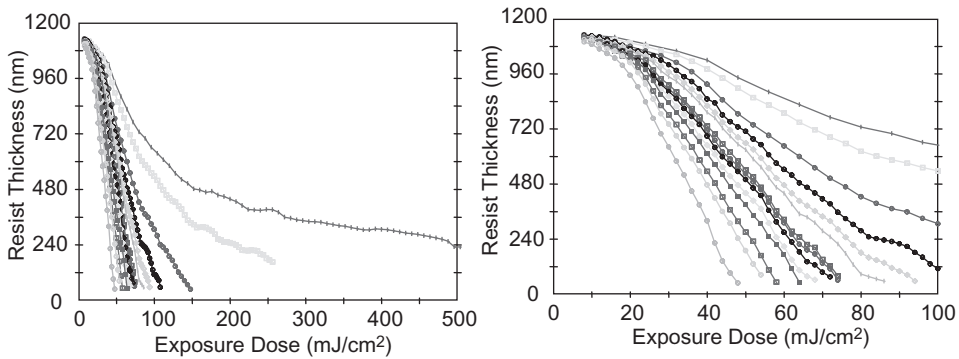


Figure 7.21 Measured contrast curves¹⁷ for an *i*-line resist at development times ranging from 9 to 201 seconds (shown here over two different exposure scales)

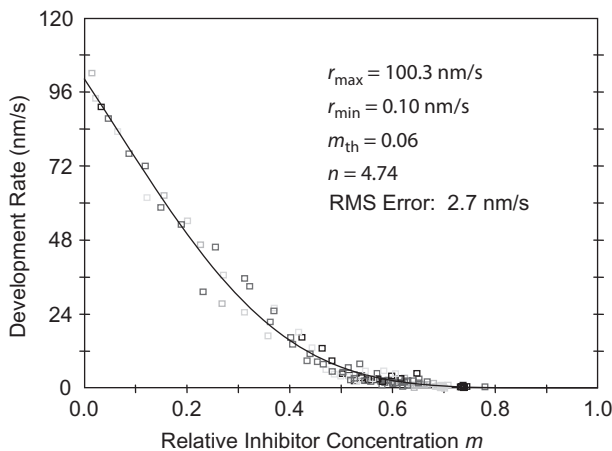


Figure 7.22 Analysis of the contrast curves generates an $r(m)$ data set, which was then fit to the original kinetic development model (best fit is shown as the solid line)

Problems

- 7.1. For the original kinetic model of Equation (7.10), what is the value of the development rate at $m = m_{th}$? For $m_{th} > 0$, what is the limit of this value as n becomes large?
- 7.2. Derive Equation (7.9).
- 7.3. Using the same approach that was used for the original kinetic development model, add the effect of developer diffusion to the surface of the resist to the enhanced kinetic model. In other words, let Equation (7.12) represent the surface reaction rate and assume that r_{resin} is proportional to D_s .
- 7.4. Consider a conventional positive resist that obeys the reaction-controlled version of the original kinetic development model of Equation (7.11). Further assume no bleaching, but absorption so that the inhibitor concentration m can be related to incident dose E and depth into resist z by

$$m = e^{-CEI_r(z)} \quad \text{where} \quad I_r(z) = e^{-\alpha z}$$

Using a spreadsheet or similar program, plot development rate versus dose for $r_{max} = 200 \text{ nm/s}$, $r_{min} = 0.05 \text{ nm/s}$, $n = 4$, $C = 0.02 \text{ cm}^2/\text{mJ}$, $\alpha = 0.5 \mu\text{m}^{-1}$, and a depth into the resist of 500 nm. Compare these results to the approximate Equation (7.19):

$$r(z) \approx r(z=0)e^{-\alpha n z}$$

Repeat for $\alpha = 0.1 \mu\text{m}^{-1}$ and $\alpha = 1.5 \mu\text{m}^{-1}$. What can you conclude about the accuracy of the approximate expression?

- 7.5. Consider a conventional positive resist that obeys the reaction-controlled version of the original kinetic development model of Equation (7.11). Further, assume no bleaching so that the inhibitor concentration m can be related to dose E by

$$m = e^{-CE}$$

Assuming that the minimum development rate r_{min} can be ignored over the dose range of interest and ignoring any variation in development rate with depth into the resist, derive an expression for the resist characteristic curve (such as that pictured in Figure 7.11). Plot the resulting equation for $r_{max} = 200 \text{ nm/s}$, $n = 7$, $C = 0.02 \text{ cm}^2/\text{mJ}$, an initial resist thickness of 800 nm, and a development time of 60 seconds. What is the value of the dose-to-clear for this case?

- 7.6. Given a conventionally measured resist characteristic curve, estimate how a small change in resist thickness would change the dose-to-clear.
- 7.7. Derive Equation (7.32).
- 7.8. Derive Equation (7.63).
- 7.9. For the Lumped Parameter Model, consider the case where only absorption causes a variation of development rate with z . In the limit of very large contrast, what is D_{eff} for the two cases of $\alpha = 0$ and $\alpha > 0$?
- 7.10. The assumption of a segmented development path can lead to a calculation of the resulting CD of a space for a given development rate function $r(x,z)$ and a given development time. How will this CD compare to the CD calculated by rigorously solving the Euler–Lagrange equation for the same conditions?

- 7.11. Use the Lumped Parameter Model to calculate CD as a function of dose for the case of a space that has a constant log-slope aerial image:

$$I(x) = I_0 e^{-s|x|}$$

where s is positive. Also assume that $x_0 = 0$ and that there is no z -dependence to the development rate. Compare this result to Equation (7.76).

- 7.12. For the Lumped Parameter Model, consider the case where only absorption causes a variation of development rate with z . Derive an expression that compares the value of resist contrast as measured with a standard photoresist Hurter–Driffield (characteristic) curve to the theoretical (LPM) contrast.

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